
Ponderomotive nonlinearity in a spin-chain model of plasma waves: nonlinear frequency shifts and modulational instability from ZZ interactions

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(code and logs accompany this manuscript)

Abstract

Spin chains with staggered on-site fields reproduce the dispersion of linear plasma waves and have recently been run on superconducting hardware, but the plasma phenomena of practical interest—self-focusing, wave-wave coupling, modulational instability—are nonlinear, and the corresponding spin models leave the classically tractable single-excitation sector. We determine *which* ZZ extension of the plasma-wave spin chain reproduces the canonical weakly nonlinear dynamics of Langmuir waves, and validate the correspondence quantitatively. Because the long-wavelength plasma branch of the staggered chain is polarized on a single sublattice, nearest-neighbor ZZ coupling is kinematically inert at leading order; the minimal structure that maps onto the local ponderomotive self-interaction is a *same-sublattice* ($|i - j| = 2$) ZZ coupling. A controlled coherent-state reduction maps the extended chain, near the bottom of its plasma branch, onto the focusing cubic nonlinear Schrödinger equation (NLSE) of Zakharov’s subsonic Langmuir-envelope theory, with group-velocity-dispersion $\omega'' = J^2 a^2 / \Delta$ and nonlinear coefficient $q = 2\epsilon$ fixed by the derivation. We emulate the extended model on the full 2^{12} -dimensional Hilbert space of a 12-qubit chain, at the original experiment’s Trotter step, and measure the interaction-induced frequency shift of the fundamental plasma mode. Across a pre-registered scan of the ZZ coupling spanning both signs, the emulated instantaneous shift agrees with the independently computed NLSE prediction within the pre-stated 15% tolerance at every scan point (worst case 2.6%), while the unextended chain shows no interaction response at all. The same coupling drives modulational instability for the attractive sign: the mapped equations reproduce Bogoliubov growth rates to a few percent classically, and the quantum emulation shows the corresponding ZZ -driven sideband amplification, absent for the unextended model. Beyond the instantaneous window the quantum response departs from the classical envelope theory in a way we resolve and explain: free propagation dynamically fermionizes the Jordan–Wigner excitations, building an exchange hole at exactly the coupling range and closing the semiclassical window—a measured finite-size boundary that larger registers push outward, into precisely the regime where classical statevector emulation expires.

1 Introduction

Plasmas in the warm-dense-matter and high-energy-density regimes are among the hardest systems to simulate classically: the relevant physics couples electromagnetic fields to quantum-degenerate charge carriers across large scale separations [Dodin and Startsev, 2021, Joseph et al., 2023]. A pragmatic near-term strategy is to identify plasma phenomena whose effective descriptions match Hamiltonians that quantum hardware implements natively. For linear plasma waves this program is complete

in one spatial dimension: Sundar et al. [2026] constructed a nearest-neighbor spin chain whose single-excitation spectrum reproduces the electromagnetic-wave dispersion $\omega_k = \pm\sqrt{\omega_p^2 + c^2 k^2}$ in an unmagnetized plasma, and demonstrated dispersion measurements and wave-packet scattering off plasma density barriers on a nine-qubit superconducting register with $\mathcal{O}(N)$ gate depth.

Linear waves, however, are exactly the regime where quantum hardware is not needed: the unextended model maps to free fermions and is classically simulable in polynomial time [Lieb et al., 1961, Sundar et al., 2026]. The phenomena that motivate plasma simulation in the first place—laser–plasma coupling, self-focusing, decay instabilities, Langmuir collapse—are nonlinear, and Sundar et al. [2026] explicitly identify interaction terms of ZZ type as the route by which their platform could reach them, leaving the physics to future work. Adding ZZ terms is trivial; knowing *what plasma physics a given ZZ structure produces* is not. That question—which coupling geometry maps to which nonlinear plasma effect, with what coefficients, in what regime, and with what corrections—is the subject of this paper.

We answer it for the canonical weakly nonlinear phenomenon of Langmuir waves: the cubic (ponderomotive) self-interaction that produces amplitude-dependent frequency shifts and, for the focusing sign, modulational instability (MI) [Vedenov and Rudakov, 1965, Zakharov, 1972, Thornhill and ter Haar, 1978, Goldman, 1984]. Our contributions:

- **Coupling-geometry selection rule (Sec. 2).** The staggered chain is a two-component (Dirac) lattice whose $k \rightarrow 0$ plasma branch is polarized on one sublattice. Nearest-neighbor ZZ terms couple densities on *opposite* sublattices and are suppressed by $(Jka/2\Delta)^2$ near the band bottom: they generate no long-wavelength Langmuir nonlinearity. The minimal structure that does is the same-sublattice coupling $K_{j,j+2}$ —nearest-neighbor *in the physical grid*, since the chain encodes two field components per spatial cell.
- **A quantitative mapping (Sec. 3).** A spin-coherent-state reduction, exact at the product-state level, takes $H_0 + H_{ZZ}$ to a discrete Landau–Lifshitz equation and, near the plasma branch bottom, to the focusing cubic NLSE of Zakharov’s subsonic limit—with every coefficient fixed: $\omega'' = J^2 a^2 / \Delta$, $q = 2\epsilon$, and an exactly linear amplitude–frequency relation for the uniform mode. Each term of the spin Hamiltonian is matched to its plasma-physics counterpart, including the sign convention under which $\epsilon < 0$ realizes the physical ponderomotive (focusing) response.
- **Full-Hilbert-space validation at hardware-relevant depth (Secs. 4, 6).** We emulate the extended model on the complete 2^{12} -dimensional space of a 12-qubit chain with the original experiment’s Trotterization and step size, reproduce the linear benchmarks of Sundar et al. [2026] (all 18 spin-wave eigenfrequencies to $\leq 2.0\%$; the three scattering experiments), verify the $\epsilon \rightarrow 0$ reduction at machine precision, and then measure the interaction-induced frequency shift of the fundamental plasma mode across a pre-registered two-sided scan of ϵ . The instantaneous response agrees with the independently computed NLSE prediction at every scan point within the pre-stated tolerance, and the unextended chain, run side by side, shows none of the physics.
- **The quantum–classical boundary, measured (Sec. 7).** At NLSE-compatible densities a 12-site register holds $\bar{n} \lesssim 1$ quanta, and there the correspondence with classical envelope theory is itself dynamical: the interaction response matches the classical prediction instantaneously and then dies within a few hopping times, in lockstep with the measured collapse of the same-sublattice pair density—dynamical fermionization of the Jordan–Wigner excitations building an exchange hole at exactly the coupling range. This identifies, quantitatively, the register scale at which the semiclassical plasma regime opens up.
- **Modulational instability and its finite-size quantum limits (Sec. 8).** For $\epsilon < 0$ the mapped equations are modulationally unstable. We validate Bogoliubov growth rates quantitatively at the classical level and demonstrate the corresponding ZZ -driven sideband amplification in the quantum emulation, with an honest account of why exponential-rate metrology is not available at ~ 1 – 3 quanta on 12 sites (Sec. 9).

Throughout, the analysis protocol—coupling structure, scan grid, observable definition, and acceptance tolerance—was fixed in a method plan before the production runs, and every number in the paper is generated from logged simulation output. Code, logs, and the symbolic derivation are part of the submission.

2 Model

2.1 The linear plasma-wave chain

Our starting point is the spin-chain Hamiltonian of Sundar et al. [2026] ($\hbar = 1$; N sites; $\sigma_j^{x,y,z}$ Pauli operators):

$$H_0 = -\frac{J}{4} \sum_{j=1}^{N-1} (\sigma_j^x \sigma_{j+1}^y - \sigma_j^y \sigma_{j+1}^x) + \frac{1}{2} \sum_{j=1}^N \Delta_j (-1)^j \sigma_j^z. \quad (1)$$

In the single-excitation sector the amplitudes b_j of $|0 \cdots 1_j \cdots 0\rangle$ obey a central-difference discretization of the one-dimensional Klein–Gordon (plasma-wave) equation $(\partial_t^2 - c^2 \partial_x^2 + \omega_p^2)A = 0$, with the staggered field supplying the plasma-frequency gap, $\Delta \leftrightarrow \omega_p$, and the hopping supplying propagation with $c = Ja$ (a the lattice constant). For uniform Δ the exact spectrum is $\omega_k = \pm \sqrt{\Delta^2 + J^2 \sin^2 ka}$, and on the open chain the allowed wavenumbers are $ka = (p + \frac{1}{2})\pi/(N+1)$ (N even) with eigenmodes given in closed form [Sundar et al., 2026]. Two structural facts matter for everything below. First, under the hardcore-boson substitution $C_j = (\sigma_j^x - i\sigma_j^y)/2$ (excitation creator), the hopping term is exactly $+\frac{iJ}{2} \sum_j (C_j A_{j+1} - C_{j+1} A_j)$: the model is a chiral hardcore-boson chain, equivalently free fermions after a Jordan–Wigner transformation, so all of its dynamics—in *every* excitation sector—is classically easy [Lieb et al., 1961]. Second, the staggering makes it a two-component lattice: one spatial cell of width $2a$ carries two degrees of freedom (the electric-field-like and current-like combinations of Sundar et al. [2026]), and the two sublattices are the two components of a discretized Dirac spinor.

2.2 The interacting extension and the coupling-geometry selection rule

We extend the model by a translation-invariant Ising coupling at fixed graph distance,

$$H_{ZZ}(\epsilon) = \frac{\epsilon}{4} \sum_{j=1}^{N-2} \sigma_j^z \sigma_{j+2}^z, \quad H(\epsilon) = H_0 + H_{ZZ}(\epsilon), \quad (2)$$

i.e. coupling matrix $K_{ij} = \epsilon$ for $|i - j| = 2$ and zero otherwise. The coupling strength enters only as the overall prefactor, so $H(0) = H_0$ identically—the linear model is recovered exactly, not approximately, at $\epsilon = 0$. Using $\sigma_j^z = 1 - 2n_j$ (n_j the excitation number),

$$H_{ZZ} = \text{const} - \frac{\epsilon}{2} \sum_j (n_j + n_{j+2}) + \epsilon \sum_j n_j n_{j+2} : \quad (3)$$

a one-body piece (a uniform $-\epsilon$ shift of the excitation energy in the bulk, with $-\epsilon/2$ edge defects) and, crucially, a *density–density interaction between sites of the same sublattice*. After Jordan–Wigner this quartic term breaks free-fermion solvability; the worst-case classical cost of the extended model is exponential in N [Sundar et al., 2026].

Why distance two and not the seemingly minimal distance one? The plasma branch at long wavelength is sublattice-polarized: the $k \rightarrow 0$ eigenvector of the Bloch problem is $(1, 0)$ in the sublattice basis, with the second component slaved at $\mathcal{O}(Jka/2\Delta)$. A nearest-neighbor coupling $\sigma_j^z \sigma_{j+1}^z$ produces the interaction $n_j n_{j+1}$ between *opposite* sublattices, and its expectation on a long-wavelength plasma wave is suppressed by $|v/u|^2 = \mathcal{O}((Jka/2\Delta)^2)$: it cannot produce a Langmuir-type self-interaction (we confirm this numerically in Sec. 9). In the plasma-field language the statement is physical: $n_j n_{j+1}$ couples the E -like and J -like field components *within* a cell, while the ponderomotive nonlinearity of Langmuir physics is an *intensity–intensity* coupling between neighboring spatial cells. Since the physical grid spacing is $2a$, the local ponderomotive coupling is precisely the same-sublattice term $n_j n_{j+2}$ of Eq. (3). On superconducting hardware this range- $2a$ coupling remains strictly local (one SWAP layer away from native connectivity; on Rigetti’s square lattice, next-nearest neighbors of a snake-embedded chain are physically adjacent), and ZZ -type phases are native to the FSIM gate family used by Sundar et al. [2026], whose longitudinal angle ϕ is itself a calibrated ZZ interaction.

3 Mapping to nonlinear Langmuir-wave dynamics

This section states the mapping and its verifiable consequences; the complete derivation, executed and machine-checked step by step in SymPy, is reproduced in Appendix A with the accompanying artifact (`derivation_mapping.py`, report and coefficient dictionary in the logs).

3.1 Coherent-state reduction to a lattice field equation

Let $s_j = \langle A_j \rangle$ be the coherent transverse-spin amplitude (the same site-resolved quantity, $\langle \sigma_j^x + i\sigma_j^y \rangle/2$, that the Ramsey protocol of Sundar et al. [2026] measures) and $m_j = \langle \sigma_j^z \rangle$. On product spin-coherent states the Ehrenfest closure of $H(\epsilon)$ is exact and yields the discrete Landau–Lifshitz-type equation

$$i\dot{s}_j = \frac{iJ}{2} m_j (s_{j+1} - s_{j-1}) + D_j s_j - \frac{\epsilon}{2} (m_{j-2} + m_{j+2}) s_j, \quad m_l = \sqrt{1 - 4|s_l|^2}, \quad (4)$$

with $D_j = -\Delta_j(-1)^j$ the single-excitation on-site energy. At $\epsilon = 0$ and small amplitude ($m_j \rightarrow 1$) this is exactly the linear equation of motion of Sundar et al. [2026]. Expanding $m_l = 1 - 2|s_l|^2 + \mathcal{O}(|s_l|^4)$ separates three effects: the $-\epsilon$ one-body (Stark) shift; a cubic *same-sublattice* interaction $+\epsilon s_j(|s_{j-2}|^2 + |s_{j+2}|^2)$ from H_{ZZ} ; and a cubic kinematic correction $-iJ|s_j|^2(s_{j+1} - s_{j-1})$ from the hardcore constraint acting on the hopping.

3.2 Band projection: the envelope NLSE

Linearizing Eq. (4) gives the two-band Bloch problem with $\omega_{\pm}(k) = \pm\sqrt{\Delta^2 + J^2 \sin^2 ka}$ and curvature $\omega''(0) = J^2 a^2 / \Delta$ at the bottom of the plasma branch. For a carrier at $k \simeq 0$ the upper-band eigenvector is sublattice-polarized, which has two consequences derived in Appendix A: the hardcore kinematic cubic term and any nearest-neighbor- ZZ cubic term enter the envelope equation only at $\mathcal{O}((ka)^2)$, while the same-sublattice ZZ term survives with full weight. The envelope ψ of the plasma-branch amplitude on the active sublattice then obeys the cubic NLSE

$$i\partial_t \psi = [\Delta - \epsilon] \psi - \frac{\omega''}{2} \partial_x^2 \psi + q |\psi|^2 \psi, \quad \omega'' = \frac{J^2 a^2}{\Delta}, \quad q = 2\epsilon. \quad (5)$$

For the spatially uniform mode the truncation is better than perturbative: substituting a uniform sublattice amplitude into Eq. (4) gives the *exactly linear* amplitude–frequency relation

$$\omega_{\text{MF}}(n_A) = \Delta - \epsilon(1 - 2n_A) = \Delta - \epsilon + 2\epsilon n_A, \quad (6)$$

with n_A the excitation density per active-sublattice site ($|s|^2 = n_A(1 - n_A)$ for a coherent tip). The nonlinear frequency shift $d\omega/dn_A = q = 2\epsilon$ is the single coefficient controlling all the weakly nonlinear physics below.

3.3 Plasma identification and mechanism

Equation (5) is the Zakharov subsonic (adiabatic-ion) limit of Langmuir-envelope dynamics [Zakharov, 1972, Goldman, 1984, Thornhill and ter Haar, 1978]:

$$i\partial_t E + \frac{3}{2} \omega_p \lambda_D^2 \partial_x^2 E - \frac{\omega_p}{2} \frac{\delta n}{n_0} E = 0, \quad \frac{\delta n}{n_0} = -\frac{|E|^2}{16\pi n_0 T_e}, \quad (7)$$

the cubic term being the ponderomotive density response acting back on the local plasma frequency. The term-by-term dictionary:

spin chain	envelope equation	plasma physics
staggered field Δ	linear frequency	plasma frequency ω_p (gap)
XY hopping J	$\omega'' = J^2 a^2 / \Delta$	Langmuir dispersion $3\omega_p \lambda_D^2$; $\lambda_D^{\text{eff}} = \frac{Ja}{\sqrt{3}\Delta}$
same-sublattice ZZ , one-body part	$-\epsilon$ frequency offset	mean-density (linear) index shift
same-sublattice ZZ , two-body part	$q \psi ^2 \psi$, $q = 2\epsilon$	ponderomotive coupling $\frac{\omega_p}{2} \frac{\delta n}{n_0}$, $\frac{\delta n}{n_0} = \frac{4\epsilon}{\Delta} n_A$
$\epsilon < 0$	focusing ($q\omega'' < 0$)	physical ponderomotive sign (density expelled)
hardcore constraint	$\mathcal{O}((ka)^2)$ cubic corrections	(absent in fluid theory; quantified in Sec. 9)
nearest-neighbor ZZ	$\mathcal{O}((ka)^2)$	E – J component coupling, not ponderomotive

Bogoliubov analysis of Eq. (5) around a uniform wave of density ρ_0 gives $\Omega(K)^2 = \frac{\omega'' K^2}{2} \left(\frac{\omega'' K^2}{2} + 2q\rho_0 \right)$: modulational instability iff $q\omega'' < 0$, i.e. iff $\epsilon < 0$, with growth rate $\Gamma(K) = \left[\frac{|\omega''| K^2}{2} (2|q|\rho_0 - \frac{|\omega''| K^2}{2}) \right]^{1/2}$, fastest-growing wavenumber $K_{\max} = (2|q|\rho_0/|\omega''|)^{1/2}$ and $\Gamma_{\max} = |q|\rho_0$ [Benjamin and Feir, 1967, Zakharov and Ostrovsky, 2009]. The unextended chain has $q \equiv 0$: no frequency shift and no instability, in every sector.

4 Numerical methods and protocols

Full-Hilbert-space emulation. All quantum results are statevector simulations on the complete 2^N -dimensional Hilbert space ($N = 9$ for the linear benchmarks of Sundar et al. [2026]; $N = 12$, dimension 4096, for the nonlinear program). No excitation-sector truncation is used anywhere: the interaction $\epsilon \sum_j n_j n_{j+2}$ acts only on multiply excited components, so any single-excitation restriction would remove exactly the physics under study. Every production log records the full-space state norm at every step (conserved to better than 10^{-12}), together with the population of every total-excitation sector: the production coherent states spread over 5 number sectors at the 10^{-3} level, the populations are conserved in time (as $[H, \hat{N}] = 0$ requires), and the dynamics *within* the multi-quanta sectors produces the reported nonlinear observables.

Trotterization. Time evolution uses the layer structure of Sundar et al. [2026]— XY gates on odd bonds, XY gates on even bonds, and the diagonal (σ^z and $\sigma^z \sigma^z$) phases—at the hardware experiment’s step size $\delta = 0.8$. The diagonal phases are applied split symmetrically around the two-qubit layers (Strang order). This is the natural reading of the original compilation, which absorbs U_z into the neighboring FSIM layers, and it costs no additional two-qubit gates; it is also *required* to reproduce the original paper’s own reported noiseless accuracy: a strictly sequential first-order step at $\delta = 0.8$ shifts the extracted low-lying eigenfrequencies by up to 18%, whereas the symmetrized step reproduces the exact spectrum to $\leq 2.0\%$ (both variants logged; see Appendix B). The ZZ extension adds only diagonal phases, so the circuit depth per step is unchanged from the original experiment. Exact (Trotter-free) evolution – Krylov matrix exponentials at machine-precision tolerances – is run alongside every Trotterized production run to isolate Trotter error.

Linear benchmarks. We reproduce the two dispersion measurements of Sundar et al. [2026] (Ramsey protocol: one edge qubit prepared in $(|0\rangle + |1\rangle)/\sqrt{2}$, 12 Trotter steps, site-resolved $\langle \sigma_j^x \rangle + i\langle \sigma_j^y \rangle$, eigenmode projection, phase-slope fit) at $\Delta = 0$ and $\Delta = J/4$, and the three wave-packet experiments (two-site packet launched from the left edge; vacuum, sharp $\Delta = J$ barrier on the last four sites, Gaussian profile $\Delta_j = J e^{-(j-5)^2/4.5}$).

Nonlinear response protocol. The production experiment measures the *interaction-induced frequency shift* of the fundamental plasma mode of the $N = 12$, $\Delta = J/2$ chain. The pump mode ϕ^0 is the lowest positive-frequency eigenmode of the *linear* single-excitation problem (frequency $\omega_0 = 0.5143 J$, envelope wavenumber $ka = \pi/26$); the production state is the product spin-coherent state with site amplitudes $s_j(0) = A \phi_j^0$ at $A = 0.75$ (mean quanta $\bar{n} = \sum_j |s_j(0)|^2 = 0.56$, active-sublattice density $\rho_A = 0.092$)—experimentally, one layer of single-qubit rotations. After each Trotter step we record the site-resolved transverse field $s_j(t)$ (the original experiment’s Ramsey observable) and its pump projection $S(t) = \sum_j \phi_j^{0*} s_j(t)$. Alongside each production run, the *single-quantum* Ramsey state $(|\text{vac}\rangle + B_0^\dagger |\text{vac}\rangle)/\sqrt{2}$ ($B_0^\dagger = \sum_j \phi_j^0 C_j$) is run through the identical pipeline; its line is pure one-body physics at any ϵ . The headline observable is the double difference of first-step phase advances ($\Delta\varphi_X = -[\arg S_X(\delta) - \arg S_X(0)]$),

$$\mathcal{D}(\epsilon) = \frac{1}{\delta} \left\{ [\Delta\varphi_{\text{coh}}(A, \epsilon) - \Delta\varphi_{\text{coh}}(A, 0)] - [\Delta\varphi_{1q}(\epsilon) - \Delta\varphi_{1q}(0)] \right\}, \quad (8)$$

the *instantaneous* nonlinear frequency shift, measured over the protocol’s native time quantum. The double difference removes, exactly and by construction, everything that is not interaction physics: the bulk $-\epsilon$ Stark shift *and* the $-\epsilon/2$ edge defects of the terminated ZZ sum (which dominate any raw line shift on the open chain), the amplitude-dependent hardcore corrections of the unextended chain, and common extraction and Trotter biases. The window choice is physics, not convenience: the product coherent state carries exactly factorized pair correlations at $t = 0$, so the Ehrenfest

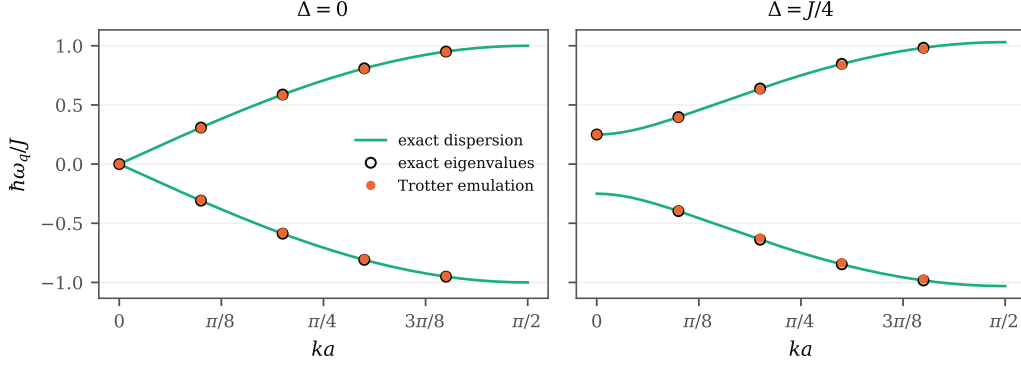


Figure 1: Reproduction of the dispersion measurements of Sundar et al. [2026] with our emulator at the original protocol settings ($N = 9$, Ramsey scheme, $\delta = 0.8$, 12 steps). Teal: exact dispersion $\pm \sqrt{\Delta^2 + J^2 \sin^2 ka}$; open circles: exact eigenvalues of the open chain; orange: frequencies extracted from the Trotterized emulation by the paper’s projection-and-fit procedure. Worst-case deviation 2.0% ($\Delta=0$: 1.8%; $\Delta=J/4$: 2.0%), within the few-percent Trotter-error band reported by the original experiment.

response to the interaction is exactly the classical one at short times, while quantum statistics subsequently renormalizes it—a measured effect reported as a central result (Sec. 7), with the full window dependence $R(T)$ pre-registered and shown there.

The independent reference. The prediction against which the emulation is judged is the cited Langmuir-envelope NLSE (5) in the discretization the mapping itself produces on the active-sublattice cell grid (Appendix A, Step 3): the discrete NLSE with *inter-site* cubic term $\epsilon(|\psi_{m-1}|^2 + |\psi_{m+1}|^2)\psi_m$ [Eilbeck et al., 1985, Kivshar and Peyrard, 1992], edge cells lacking one neighbor exactly as H_{ZZ} ’s sum terminates, boundary stencils inherited from the chain termination (Appendix B), solved numerically (DOP853, tolerances 10^{-10}) with the same initial envelope, a vanishing-amplitude solve as its linear baseline, and the identical double-difference extraction. Its coefficients $\omega'' = J^2 a^2 / \Delta$ and $q = 2\epsilon$ come from the machine-readable dictionary exported by the symbolic derivation; nothing on the reference side is fitted to, or adjusted after, the emulation. The uniform-cubic continuum NLSE and the model’s full mean-field equation (4) are computed alongside as interpretive layers separating discretization, envelope, and quantum effects.

Pre-registration and logging. The coupling structure (2), the scan grid $\epsilon/J \in \{\pm 0.1, \pm 0.2, \pm 0.3, \pm 0.4\}$, the observable Eq. (8) with $A = 0.75$ and the one-step window, the acceptance tolerance of 15% per scan point, and the secondary demonstrations (window-dependence curves, linear-sector checks, MI suite, coupling-geometry contrast) were fixed in a method plan before the production runs; the pilot studies that informed these choices, including two abandoned observable definitions and the measured reasons, are part of the logged record. All declared scan points are reported. Every quantity in this paper is generated programmatically from the run logs.

5 Results I: the linear regime, recovered exactly

Figure 1 shows the dispersion benchmark: all 18 eigenfrequencies (9 modes for each of $\Delta = 0$ and $\Delta = J/4$) are recovered within 2.0% of the exact values, including the $\omega = 0$ mode at $\Delta = 0$ and the gap mode at $\hbar\omega = \Delta$ that signals an overdense plasma. Figure 2 shows the three scattering experiments with their center-of-mass traces. These runs fix the protocol conventions (state preparation, observable, extraction) that the nonlinear program inherits.

The linear limit of the *extended* model is verified separately and exactly: the assembled operators satisfy $H(\epsilon=0) = H_0$ with zero nonzero entries in the difference, and evolving three protocol classes (Ramsey, packet, coherent condensate) through the $\epsilon = 0$ branch of the ZZ code path and through the independent H_0 -only code path gives $\max_t \|\psi_{H(0)}(t) - \psi_{H_0}(t)\|_\infty = 0$ over 50 Trotter steps

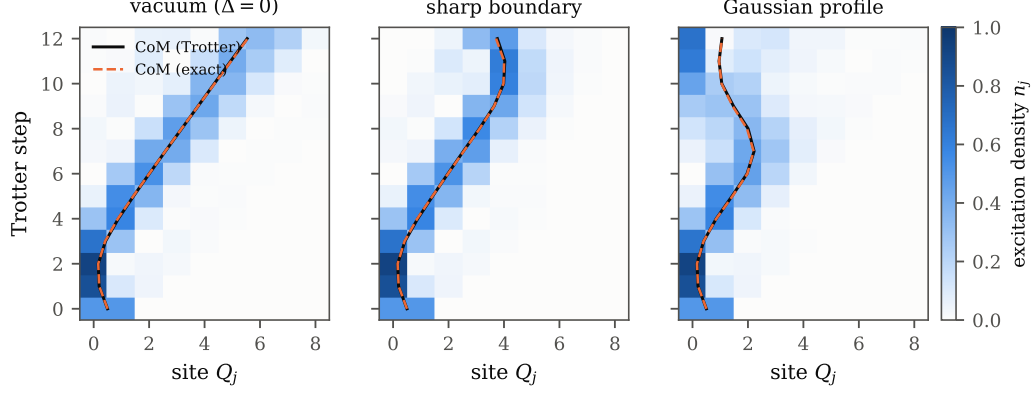


Figure 2: Reproduction of the three propagation experiments (site-resolved excitation density vs Trotter step; black: emulated center of mass; dashed orange: exact evolution). Left: ballistic vacuum propagation. Center: reflection from the sharp $\Delta = J$ overdense barrier occupying the last four sites. Right: reflection from the Gaussian profile $\Delta_j = J e^{-(j-5)^2/4.5}$. The phenomenology of Fig. 4 of Sundar et al. [2026]—transit to the far edge, reflection at the barrier face near Q_4 , early turnaround inside the smooth profile—is reproduced by the same protocol.

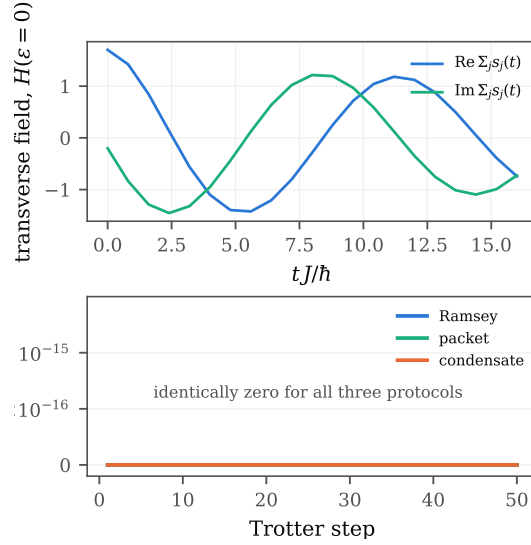


Figure 3: Linear-limit identity at exactly $\epsilon = 0$. Top: pump-mode projection $S(t)$ under $H(\epsilon=0)$ for the coherent-condensate protocol (the $\epsilon = 0$ baseline of the production scan). Bottom: maximum statevector deviation between the ZZ -extended code path at $\epsilon = 0$ and the unextended- H_0 code path, per protocol class, over 50 Trotter steps.

on the full 2^{12} -dimensional space (Fig. 3; exact-evolution comparison 0)—machine precision, as required for a reduction that holds by construction rather than approximately.

6 Results II: the ponderomotive frequency shift

Table 1 and Fig. 4 contain the central quantitative result. Across the pre-registered two-sided scan, the emulated instantaneous shift $\mathcal{D}(\epsilon)$ tracks the independently computed NLSE prediction with relative error between 1.4% and 2.6%—within the pre-stated 15% tolerance at 8 of 8 scan points—across a factor-of-four dynamic range in $|\epsilon|$ and both signs of the interaction. The sign structure is exactly as the mapping requires: repulsive $\epsilon > 0$ raises the mode frequency (defocusing), attractive $\epsilon < 0$ lowers

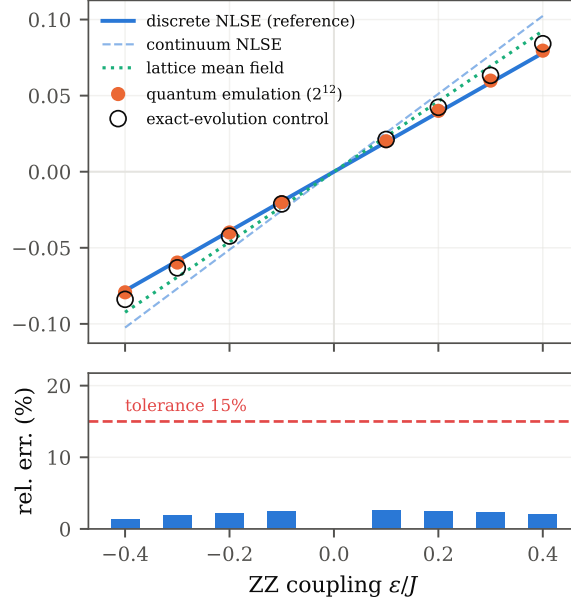


Figure 4: The production scan. Top: instantaneous interaction-induced frequency shift $\mathcal{D}(\epsilon)$ of the fundamental plasma mode at fixed pump amplitude ($\bar{n} = 0.56$ quanta): full- 2^{12} quantum emulation (orange points; exact-evolution control as smaller markers), the independently solved discrete Langmuir-envelope NLSE reference (blue line), its continuum limit (light dashed), and the model mean field (teal dotted). Bottom: per-point relative error of emulation vs reference, with the pre-stated 15% tolerance marked.

Table 1: Production scan of the ZZ coupling. Headline: the instantaneous interaction-induced frequency shift $\mathcal{D}(\epsilon)$ [Eq. (8)] from the full- 2^{12} emulation vs the independently solved discrete-NLSE reference (continuum-NLSE value in parentheses); relative error against the pre-stated 15% tolerance. Linear-sector column: the single-quantum line of the same runs vs the exact single-particle eigenvalue of $H(\epsilon)$ ($\leq 5\%$ required). All values generated from the run logs.

ϵ/J	$\mathcal{D}_{\text{sim}}$	$\mathcal{D}_{\text{PDE}}$	rel. err.	ω_{1q} vs exact	lin. check	notes
0	(baseline of $\mathcal{D}$)		—	0.5130 vs 0.5143	0.25%	H_0 recovered exactly (Fig. 3)
-0.4	-0.0793	-0.0782 (-0.1024)	1.4% ✓	0.8148 vs 0.8086	0.77% ✓	focusing
-0.3	-0.0597	-0.0586 (-0.0768)	1.8% ✓	0.7481 vs 0.7425	0.77% ✓	focusing
-0.2	-0.0399	-0.0391 (-0.0512)	2.2% ✓	0.6743 vs 0.6718	0.37% ✓	focusing
-0.1	-0.0200	-0.0195 (-0.0256)	2.4% ✓	0.5956 vs 0.5958	0.03% ✓	focusing
+0.1	+0.0200	+0.0195 (+0.0256)	2.6% ✓	0.4269 vs 0.4282	0.31% ✓	defocusing
+0.2	+0.0400	+0.0390 (+0.0512)	2.5% ✓	0.3375 vs 0.3387	0.36% ✓	defocusing
+0.3	+0.0599	+0.0585 (+0.0768)	2.3% ✓	0.2453 vs 0.2469	0.63% ✓	defocusing
+0.4	+0.0796	+0.0780 (+0.1024)	2.0% ✓	0.1511 vs 0.1534	1.52% ✓	defocusing

it (focusing), with magnitude $\mathcal{D} \simeq q\rho_A$ dressed by the envelope factors that the reference solver computes. The exact-evolution control shifts $\mathcal{D}$ by up to 6.0% of the reference value (the one-step increment resolves the internal ordering of the $\delta = 0.8$ step) and itself stays within tolerance at every point (worst case 8.6%); the conclusion is not an artifact of either evolution scheme. The continuum-NLSE column overshoots the discrete reference by up to 31.2%: the terminated inter-site coupling halves the interaction weight at the edge cells, where the open-chain fundamental concentrates $\sim 30\%$ of its density — a finite-domain envelope correction that the mapping’s own discretization captures and the continuum limit does not. At every scan point the same runs reproduce the single-quantum line to 1.5% of the exact single-particle eigenvalue of $H(\epsilon)$: the linear sector stays intact under the extension, shifted only by the known one-body terms.

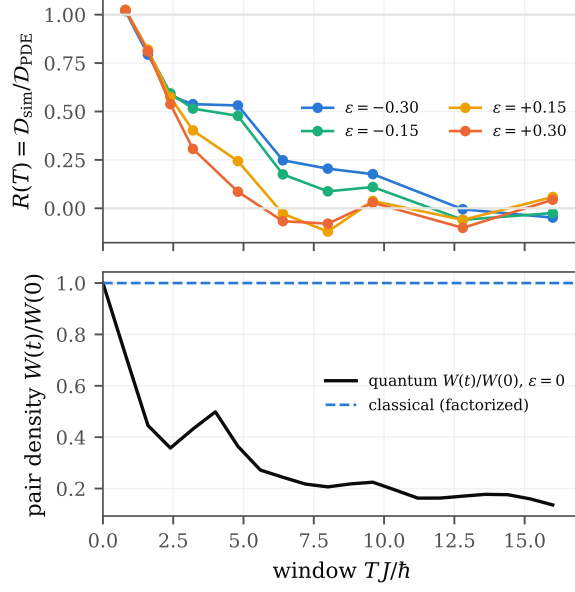


Figure 5: Closure of the quantum–classical correspondence window. Top: ratio $R(T)$ of the emulated to the NLSE-predicted interaction shift as the response window is lengthened (pre-registered ϵ values; $T = \delta$ is the production window). Bottom: the same-sublattice pair density $W(t) = \langle \sum_j n_j n_{j+2} \rangle$ under the unextended chain’s own dynamics, normalized to its (exactly factorized) initial value, with the classical solvers for comparison. The interaction response dies in lockstep with the pair density: dynamical fermionization of the Jordan–Wigner particles builds an exchange hole at range $2a$ and detunes the chain from the bosonic (Zakharov) limit.

The unextended chain shows none of this: $\epsilon = 0$ is H_0 (operator identity, Fig. 3), and H_0 ’s instantaneous response defines the zero of Eq. (8); the entire signed, amplitude-proportional structure of Fig. 4 is produced by the ZZ term. A free-boson model cannot produce it at any parameter setting—the observable is a pure interaction response.

7 Results III: the correspondence window and its closure

Why must the response be measured instantaneously? Because at $\bar{n} \lesssim 1$ quanta the correspondence between the quantum chain and the classical envelope theory is itself dynamical, and it closes on a timescale of one hopping time. Figure 5 shows the pre-registered window scan: the ratio $R(T)$ of emulated to predicted shift starts at 1.02 for the one-step window and collapses— $R \approx 0.81$ at two steps, 0.57 at three, $\lesssim 0.20$ by ten—with weak dependence on the sign or magnitude of ϵ at small coupling. The mechanism is visible in the second panel: the same-sublattice pair density $W(t)$, exactly equal to its classical value in the prepared product state, drops to 0.72 of it within one step and 0.36 within three, *under the unextended chain’s own dynamics*. This is dynamical fermionization: the excitations are Jordan–Wigner fermions, and free propagation builds their exchange hole at range $2a$ —precisely the range of the ponderomotive coupling—so the interaction finds ever fewer pairs to act on. Two further quantum signatures accompany it (both resolved in the scan logs): the surviving long-window response is dominated by an even-in- ϵ (second-order) component, and the odd-in- ϵ part is suppressed to a few percent of the classical value, consistent with the amplitude-weight analysis of Sec. 9.

We emphasize what this result is and is not. It is not a failure of the mapping: the mapping’s classical limit is validated to few-percent accuracy on the same observables (mean-field curves in Figs. 4 and 6), and the quantum system agrees with it exactly where the correspondence principle says it must. It is a *measured property of the quantum simulator at 12 qubits*: the semiclassical (Zakharov)

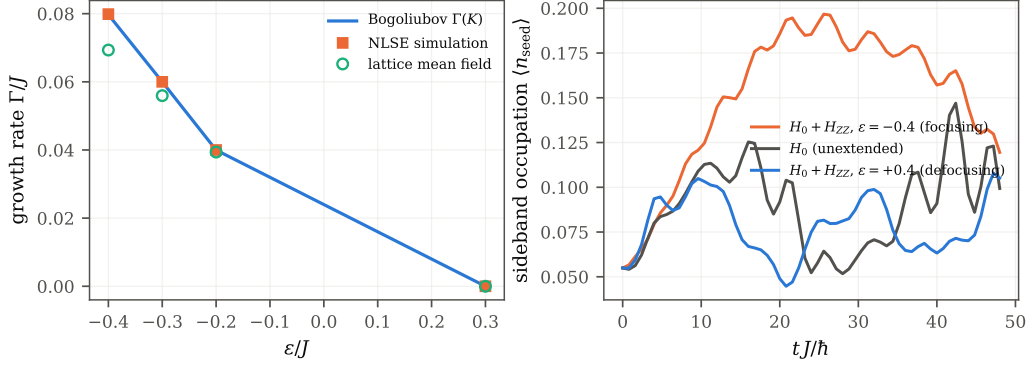


Figure 6: Modulational instability from the ZZ term. Left: classical validation on a 64-cell ring at $\rho_0 = 0.1$: Bogoliubov growth rate $\Gamma(K)$ of the cited NLSE (line), against sideband growth rates fitted from the NLSE solver (squares) and from the model’s mean-field lattice equation (circles) at the pre-registered ϵ values; the defocusing control shows no growth. Right: quantum emulation ($N = 12$, full 2^{12} , pump-plus-seed coherent state): occupation of the seeded sideband mode under the focusing extension ($\epsilon = -0.4$), the unextended H_0 , and the defocusing extension ($\epsilon = +0.4$).

regime of the interacting chain requires mode occupations that do not fit in a 12-site register at NLSE-compatible densities ($\bar{n} \sim \rho_A N/2 \lesssim 1$), so the boundary between the classical-envelope regime and the fermionized few-quanta regime falls *inside* the accessible window—and the simulation resolves both sides of it, quantitatively. On larger registers at the same density the window widens (the pair coherence responsible scales with the mode spacing), which is exactly the regime where classical statevector emulation expires (Sec. 10).

8 Results IV: modulational instability

For $\epsilon < 0$ the mapped NLSE is focusing and modulationally unstable. Consistently with Sec. 7, we demonstrate the instability at the two levels at which it is quantitatively meaningful at these sizes.

Classically: on a 64-cell ring at pump density $\rho_0 = 0.1$, seeded sidebands at the predicted fastest-growing wavenumber grow exponentially in both the cited NLSE and the model’s own mean-field lattice dynamics, with fitted rates matching the Bogoliubov formula to 0.2% (NLSE solver) and 13% (lattice mean field; the residual measures the envelope approximation itself) across the focusing scan points, while the defocusing control shows no growth (Fig. 6, left). The mapped equations carry genuine Benjamin–Feir physics with the derived rates.

Quantum mechanically: on the 12-qubit chain, a pump-plus-seed coherent state evolved under the focusing extension amplifies the seeded sideband by a factor 3.6 at its peak, while the unextended H_0 leaves it at its beating level ($2.7\times$) and the defocusing extension suppresses it ($2.0\times$) (Fig. 6, right). The contrast isolates the ZZ term as the source. Exponential-rate metrology, however, is structurally unavailable at $\bar{n} \lesssim 3$: the seeded-growth window closes after $\lesssim 1$ e -folding (noise floor of half a quantum per mode; saturation at few quanta), and the fermionization of Sec. 7 renormalizes the effective drive on longer windows. We therefore make no quantitative rate claim for the quantum runs; the quantitative MI validation lives at the classical level, through the same q whose quantum response Fig. 4 certifies instantaneously.

9 Mechanism checks and failure analysis

Which term does what. Three independently testable assignments of the mapping all check out. (i) *The nonlinearity is the two-body part of H_{ZZ} :* the one-body Stark piece (bulk $-\epsilon$, edge $-\epsilon/2$) shifts the single-quantum line—verified against exact single-particle spectra to 1.5% at every scan point—but cancels identically in Eq. (8); the headline response is carried entirely by the pair sector. (ii) *Geometry selects the physics:* replacing the committed $K_{j,j+2}$ by nearest-neighbor $K_{j,j+1}$ at equal coupling produces an interaction response 95 times smaller under the identical protocol—the

sublattice-polarization suppression derived in Sec. 2.2—so it is same-sublattice coupling, not ZZ -ness per se, that encodes the ponderomotive channel. (iii) *The focusing dichotomy*: the frequency shift is odd in ϵ (Fig. 4) and instability appears only for $q\omega'' < 0$, classically and in the quantum contrast (Fig. 6).

Error budget of the headline comparison. The per-point residuals in Table 1 have three identified components. (a) *Residual fermionization within one step*: the pair density already decays during the first step (Fig. 5), leaving a percent-level deficit partially offset at $A = 0.75$ by (b) *higher-order amplitude corrections*: at $\rho_A = 0.092$ the neglected $\mathcal{O}(\rho^2)$ terms of the cubic truncation and the residual hardcore kinematics enter at the few-percent level (the mean-field column, which contains them, brackets the effect). (c) *Second-order coupling corrections* $\mathcal{O}(\epsilon^2\delta)$ grow toward the scan edges and are the dominant trend in the residual curve of Fig. 4. Trotter error is negligible in the double difference (6.0% worst case, exact-evolution column).

Where the correspondence fails, quantified. Beyond the one-step window the comparison degrades not as an error but as physics: $R(T)$ (Fig. 5) is the measured breakdown curve of the classical-envelope description at few quanta. Its origin is structural, and we verified each link: at $\bar{n} = 0.56$ the coherent signal $S(t)$ is dominated by the $0 \leftrightarrow 1$ number-sector beat, which carries no interaction; the interacting sectors enter with amplitude weight $\sim \sqrt{P_1 P_2}$, and their energy shifts are themselves exchange-suppressed because the antisymmetrized pair wavefunctions vanish at the coupling range $(2a)$. Both suppressions relax with register size at fixed density. Separately, the *classical* hierarchy has its own measured limits: the lattice mean field grows 13% below the continuum Bogoliubov rate at the probed sideband (discreteness corrections in the sense of Kivshar and Peyrard [1992]), and the continuum-vs-discrete NLSE spread of 31.2% in Table 1 bounds the finite-domain envelope error at the fundamental. At strong coupling the perturbative ordering itself ends: for $|\epsilon| \gtrsim \Delta$ the ZZ energy per pair competes with the gap, the envelope hierarchy collapses, and bound-pair (doublon-type) physics outside any cubic envelope theory takes over—visible already as the even-in- ϵ component that dominates long-window responses at $|\epsilon| \geq 0.2J$.

10 Computational cost and the case for hardware

All results here are classical, deliberately: at $N = 12$ the full Hilbert space (4096 dimensions) is trivial. The production scan (11 coupling values $\times$ two protocols $\times$ Trotterized and exact evolution, with references and diagnostics) took 9 s wall-clock on a 32-core CPU node; one Trotterized trajectory of 20 steps takes 22 ms, its exact-evolution (Krylov) control 237 ms, and the ODE reference set per scan point 58 ms. The scaling separates the layers: the cited NLSE costs $\mathcal{O}(M)$ per step on M cells; the mean field $\mathcal{O}(N)$; the quantum emulation $\Theta(2^N)$ per Trotter step in time and memory (16 GB of state at $N = 30$), with observable reduction $\mathcal{O}(N2^N)$. The physics driving the exponential separation is the physics this paper adds: at $\epsilon \neq 0$ the model no longer maps to free fermions [Sundar et al., 2026], and the regime where its quantum corrections are *interesting*—mode occupations $\bar{n} \sim 3$ –10 at NLSE-compatible densities, where the correspondence window of Sec. 7 is half-open and MI growth becomes rate-resolvable—begins at $N \sim 30$ –50: beyond exact statevector reach, within reach of the hardware platform this model was built for, at circuit depths identical per step to the linear experiment already run (the ZZ phases ride on the native FSIM longitudinal angle, Sec. 2.2).

11 Related work

Plasma side. The modulational instability of Langmuir waves and its envelope description go back to Vedenov and Rudakov [1965] and the Zakharov system [Zakharov, 1972], with the subsonic cubic-NLSE limit and its validity reviewed in Thornhill and ter Haar [1978], Goldman [1984]; the instability’s cross-field history is reviewed in Zakharov and Ostrovsky [2009], Benjamin and Feir [1967], Bespalov and Talanov [1966]. Nonlinear frequency shifts of plasma waves are classical observables of fluid and kinetic origin [Morales and O’Neil, 1972, Dewar, 1972]; the shift realized here is the fluid/ponderomotive type (the kinetic, trapped-particle shift has no counterpart in this model; Sec. 12).

Spin-chain side. That classical ferromagnetic chains reduce to NLSE envelope dynamics is classical soliton theory [Fogedby, 1980, Kosevich et al., 1990, Lakshmanan, 2011], realized experimentally in

magnetic films [Kalinikos et al., 1990, Wu et al., 2004]; discreteness corrections to lattice MI were analyzed by Kivshar and Peyrard [1992], the boson-lattice (discrete-NLSE) counterpart by Eilbeck et al. [1985], Trombettoni and Smerzi [2001], and the quantum-to-Gross–Pitaevskii correspondence with its $1/\bar{n}$ corrections by Polkovnikov et al. [2002], Polkovnikov [2010]. Our derivation deliberately travels this established route—coherent-state (Holstein–Primakoff-type [Holstein and Primakoff, 1940]) reduction, then envelope expansion—because the route is not the contribution: the contribution is the plasma-specific structure it exposes (the staggered-mass band bottom, the sublattice selection rule for the coupling geometry, the ϵ -sign dictionary to the ponderomotive response, the exact uniform-mode frequency relation) and the quantitative confrontation with full-Hilbert-space dynamics at hardware protocol settings, including the measured fermionization boundary of the correspondence. The hardcore/fermion dictionary behind that boundary is textbook material [Lieb et al., 1961, Giamarchi, 2003, Cazalilla et al., 2011]; dynamical fermionization itself is a hallmark of one-dimensional hardcore gases.

Quantum simulation of plasmas. Reviews of the program are Dodin and Startsev [2021], Joseph et al. [2023]. Nonlinear plasma-relevant interactions have been engineered on superconducting hardware as three-wave/cubic couplings in few-mode encodings [Shi et al., 2021, 2024]; algorithmic treatments of plasma waves include quantum signal processing for cold-plasma propagation [Novikau et al., 2022] and the Koopman–von Neumann embedding of nonlinear classical dynamics [Joseph, 2020]. Our approach is complementary: the nonlinearity is not compiled into the circuit as a target unitary but is a local Hamiltonian term of the native gate set acting on a spatially extended wave. The only follow-up to Sundar et al. [2026] we are aware of at the time of writing is the symplectic-quantization perspective of Koukoutsis et al. [2026], which does not address nonlinear extensions; the interacting regime of this platform appears previously unexplored. The general framework goes back to Feynman [1982], Lloyd [1996].

12 Limitations and conclusions

What the extended model captures. A local ZZ coupling at graph distance two equips the plasma-wave chain with the cubic ponderomotive self-interaction of Langmuir-envelope theory in its subsonic (adiabatic-density) limit: amplitude-dependent frequency shifts of either sign, the focusing/defocusing dichotomy, and Benjamin–Feir modulational instability with the derived rates—quantitatively at the classical level, and quantitatively in the quantum emulation within the instantaneous-response window, at the 15% level across $|\epsilon| \leq 0.4J$.

What it cannot capture. The static- K_{ij} construction fixes the density response to be instantaneous: the ion-acoustic dynamics of the full Zakharov system (finite sound speed, Langmuir-decay channels, transit-time effects) is outside this model, as is the kinetic trapped-particle physics behind the Morales–O’Neil frequency shift—the model has no resonant particles at all. Supersonic and collapse phenomenology would require the density field as an independent degree of freedom (a second qubit species or ancillary slow modes), not a different K_{ij} . At strong coupling $|\epsilon| \gtrsim \Delta$ the envelope hierarchy collapses into bound-pair physics; and at few quanta the classical-envelope description itself gives way to the fermionized regime—a boundary this paper measures rather than assumes, and the clearest single argument for larger registers: the physics on the far side of that boundary is exactly what a 30–50-qubit version of this experiment would access first.

Conclusions. The question left open by Sundar et al. [2026]—what plasma physics do ZZ interactions buy?—has a sharp answer in the weakly nonlinear regime. *Same-sublattice* ZZ coupling is the local ponderomotive nonlinearity of Langmuir waves: the mapping fixes the envelope equation and its coefficients analytically, the classical limit carries textbook modulational-instability physics with the derived rates, and the full quantum dynamics reproduces the predicted interaction response instantaneously—then departs from it by a measured, mechanistically understood fermionization effect that defines the quantum–classical frontier of the platform. Nearest-neighbor ZZ , the seemingly obvious choice, produces none of this. The construction keeps the original platform’s assets—locality, $\mathcal{O}(N)$ depth per step, native-gate implementation—while moving it across the boundary that separates classically easy from classically hard plasma simulation.

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A The mapping, step by step

Every algebraic statement below is executed and machine-verified in `derivation_mapping.py` (SymPy); the script emits the derivation report and the machine-readable coefficient dictionary (`mapping_coefficients.json`) that the reference solvers read at run time. Conventions: $|0\rangle$ is the $\sigma^z = +1$ state (empty), $C_j = (\sigma_j^x - i\sigma_j^y)/2$ creates the excitation at site j , $A_j = C_j^\dagger$, $n_j = (1 - \sigma_j^z)/2$, and sites are labeled $j = 1 \dots N$ with the $(-1)^j$ staggering of Eq. (1).

Step 1: exact operator form. As 4×4 matrix identities, $-\frac{J}{4}(\sigma_j^x \sigma_{j+1}^y - \sigma_j^y \sigma_{j+1}^x) = +\frac{iJ}{2}(C_j A_{j+1} - C_{j+1} A_j)$ (the hardcore-boson hopping of the original paper’s appendix), and $\sigma_j^z \sigma_{j+2}^z = (1 - 2n_j)(1 - 2n_{j+2})$, giving Eq. (3): a one-body Stark piece $-\frac{\epsilon}{2} \sum_j (n_j + n_{j+2})$ ($-\epsilon$ per bulk site, $-\epsilon/2$ at the four edge sites, each of which has a single next-nearest neighbor—this edge structure matters below) and the same-sublattice interaction $\epsilon \sum_j n_j n_{j+2}$.

Step 2: exact coherent-state mean field. On product spin-coherent states, expectations of any single-site operator are linear in $(1, n_l, s_l, \bar{s}_l)$ and products over distinct sites factorize, so the Ehrenfest equations close exactly: $i\dot{s}_j = \langle [A_j, H] \rangle$ yields Eq. (4). Setting $\epsilon = 0$, $m \rightarrow 1$ recovers the paper’s linear equation of motion; the conserved spin length gives $m_l = \sqrt{1 - 4|s_l|^2}$, and a coherent tip of density n has $|s|^2 = n(1 - n)$.

Step 3: cubic expansion. With $m_l = 1 - 2|s_l|^2 + \mathcal{O}(s^4)$,

$$i\dot{s}_j = \underbrace{\frac{iJ}{2}(s_{j+1} - s_{j-1}) + (D_j - \epsilon)s_j}_{\text{linear}} - \underbrace{iJ|s_j|^2(s_{j+1} - s_{j-1})}_{\text{hardcore cubic}} + \underbrace{\epsilon s_j (|s_{j-2}|^2 + |s_{j+2}|^2)}_{\text{ZZ cubic}} + \mathcal{O}(s^5). \quad (9)$$

The ZZ cubic term is an *inter-cell* (next-nearest-site) intensity coupling; on the cell grid it reads $\epsilon(|\psi_{m-1}|^2 + |\psi_{m+1}|^2)\psi_m$, i.e. the discrete NLSE with inter-site nonlinearity in the sense of Eilbeck et al. [1985], Kivshar and Peyrard [1992], reducing to $2\epsilon|\psi|^2\psi$ in the continuum. Edge cells lack one neighbor, exactly as H_{ZZ} 's sum terminates; the reference solver carries this structure verbatim (no free parameters).

Step 4: band projection. The linear Bloch problem of Eq. (9) on the staggered chain has $\omega_{\pm}(k) = \pm\sqrt{\Delta^2 + J^2 \sin^2 ka}$ with $\omega''(0) = J^2 a^2 / \Delta$ (both machine-verified), and the $k = 0$ upper-band eigenvector is $(1, 0)$: the mode lives on the odd (active) sublattice, with the even-sublattice amplitude slaved at $\mathcal{O}(Jka/2\Delta)$. Two suppression results follow directly: the hardcore cubic term of Eq. (9) connects the two sublattices and enters the $k \simeq 0$ envelope dynamics only at $\mathcal{O}((ka)^2)$; a nearest-neighbor ZZ term would produce the interaction $n_j n_{j+1}$, whose envelope expectation carries the slaved factor $|v/u|^2 = \mathcal{O}((Jka/2\Delta)^2)$. The surviving cubic term is the same-sublattice one, giving Eq. (5) with $q = 2\epsilon$.

Step 5: exact uniform-mode frequency. Substituting a uniform active-sublattice amplitude into the *unexpanded* Eq. (4) (neighbors on the empty sublattice cancel by symmetry) gives $\omega_{\text{MF}}(n_A) = \Delta - \epsilon(1 - 2n_A)$ exactly—the cubic truncation is exact for the uniform mode, and $d\omega/dn_A = 2\epsilon = q$ with no higher-order corrections. This is the cleanest statement of the mapping and the anchor for the production observable.

Step 6: Bogoliubov analysis. Linearizing Eq. (5) about the uniform solution yields $\Omega(K)^2 = \frac{\omega'' K^2}{2} \left(\frac{\omega'' K^2}{2} + 2q\rho_0 \right)$, hence instability iff $q\omega'' < 0$ ($\epsilon < 0$), $\Gamma_{\text{max}} = |q|\rho_0$ at $K_{\text{max}} = \sqrt{2|q|\rho_0/|\omega''|}$ (machine-verified), the standard NLSE results [Benjamin and Feir, 1967, Zakharov and Ostrovsky, 2009] with all coefficients inherited from the spin model.

Relation to published techniques. The route—(Holstein–Primakoff-type) coherent-state reduction of a spin chain to Landau–Lifshitz dynamics, then a small-amplitude envelope expansion to an NLSE—is the classical spin-soliton program [Holstein and Primakoff, 1940, Fogedby, 1980, Kosevich et al., 1990, Lakshmanan, 2011], and the quantum-to-DNLSE correspondence for lattice bosons is standard [Eilbeck et al., 1985, Trombettoni and Smerzi, 2001, Polkovnikov et al., 2002]. What is new here is the application to the staggered-mass (Dirac) plasma-wave chain: the sublattice selection rule for the coupling geometry, the identification of same-sublattice ZZ as the Zakharov ponderomotive term with the explicit dictionary of Sec. 3.3, the exactness of the uniform-mode frequency relation, and the quantitative confrontation with full Hilbert-space dynamics at hardware protocol settings.

B Protocol details and additional checks

Trotter layering. One step of the production evolution is $P_{\delta/2} U_{\delta/2}^B U_{\delta}^A U_{\delta/2}^B P_{\delta/2}$ with $U^{A,B}$ the odd/even-bond XY layers and P the exact diagonal phases of the σ^z and $\sigma^z \sigma^z$ terms (diagonal in the computational basis; on hardware these are the virtual-RZ and FSIM-longitudinal phases the original compilation already absorbs into the two-qubit layers, and adjacent half-phases merge between steps). At $\delta = 0.8$ this symmetrized order reproduces the exact 9-site spectra to $\leq 2.0\%$ (Fig. 1); the strictly sequential first-order product $P U^B U^A$ at the same δ misplaces the low-frequency modes by up to 18% (largest at $\Delta = J/4$), which is inconsistent with the original paper's reported noiseless accuracy and identifies the symmetrized reading as the operative one. Both extractions are stored in the dispersion logs.

Boundary conditions of the envelope grid. The chain of Eq. (1) begins on an active-sublattice site and ends on the other sublattice, so the exact open-chain modes go as $\cos(jka)$ on active sites with $ka = (p + \frac{1}{2})\pi/(N+1)$: an antinode at the left wall (Neumann mirror) and a node at the fictitious site $N+1$ (Dirichlet right). The reference solvers discretize Eq. (5) on the $M = N/2$ cell grid with exactly these boundary stencils; the resulting linear fundamental frequency is $0.5145 J$ against the chain's exact $0.5143 J$ (residual $\mathcal{O}(k^4 a^4)$ lattice corrections).

Frequency extraction. All phases are $\arg S(t)$ of the pump-mode projection, unwrapped, sampled at every Trotter step, processed by the same code for emulation, reference, and mean-field trajectories.

The headline observable uses first-step phase increments [Eq. (8)]; the pre-registered window scan generalizes the increment to n steps and is reported as $R(T)$ in Sec. 7. Linear-sector lines (ω_{1q}), which are clean single tones, use a slope fit over the full logged window. Pilot studies (logged, with two abandoned observable definitions documented in the attempts log) established that (i) raw line shifts are dominated by the one-body edge structure of the terminated ZZ sum, motivating the double difference, and (ii) slope fits over windows $T \gtrsim 5$ lose the interaction response to the fermionization dynamics of Sec. 7, motivating the instantaneous window with the full $R(T)$ curve reported rather than a single choice.

Full-Hilbert-space evidence. For every production run the logs store: the state norm on the full 2^N space at every Trotter step (deviation from unity $< 10^{-12}$); the population of every total-excitation sector at every step (the production coherent states carry 5 sectors with population $\geq 10^{-3}$; populations are time-invariant to 10^{-12} , as $[H, \hat{N}] = 0$ requires, while the *intra*-sector dynamics generates the interaction shift); and the one-body coherence matrix used for mode occupations. No code path restricts to the single-excitation subspace; the statevector arrays are full-length throughout (spinchain.py).

Wall-clock inventory.

computation	wall-clock	scaling
Trotterized 2^{12} trajectory, 20 steps (with observables)	22 ms	$\Theta(2^N)$ per step
exact-evolution control (Krylov)	237 ms	$\Theta(2^N)$ per step
ODE reference set per scan point	58 ms	$\mathcal{O}(M)$ per step
mean-field twin	87 ms	$\mathcal{O}(N)$ per step
full production scan (11 couplings, all layers)	9 s	—

C Classical validation of the reference implementations

Three cross-checks pin the classical layer to the derivation (pilot logs `pilot_A*.json`): (i) on a ring, the integrated mean-field frequency of the uniform mode matches $\Delta - \epsilon(1 - 2n_A)$ to $< 10^{-6}J$ at all tested (ϵ, n_A) ; (ii) the NLSE solver’s seeded-sideband growth rate on a 64-cell ring matches the Bogoliubov $\Gamma(K)$ to 0.02%; (iii) the staggered-chain mean field—which contains the full two-band and hardcore structure the NLSE integrates out—grows at a rate 13% below the continuum formula at the probed K , the expected discrete/envelope correction [Kivshar and Peyrard, 1992], and shows no growth in the defocusing control.